

Elastic Load Balancer in AWS

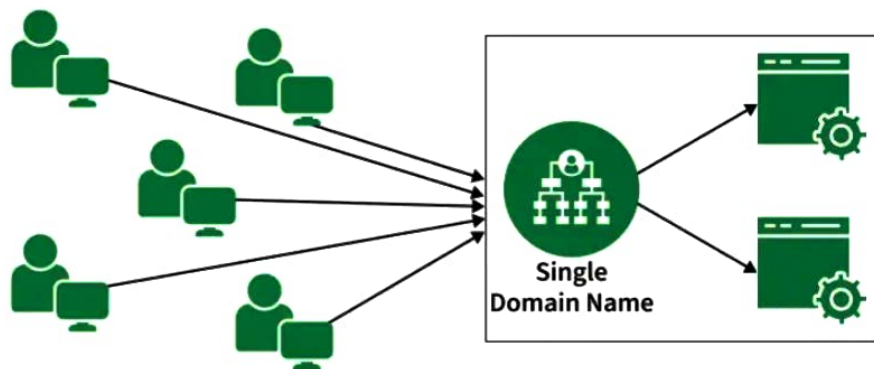
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Elastic Load Balancing (ELB) automatically distributes incoming application traffic across multiple targets such as Amazon EC2 instances, containers, IP addresses, and Lambda functions. It acts as the "front door" for your application, ensuring high availability, fault tolerance, and elasticity.

Core Concepts

- **Listeners:** A listener checks for incoming connection requests on a specific protocol and port.
- **Target Groups:** A logical group of backend resources such as EC2 instances, containers, IP addresses, or Lambda functions. The load balancer sends traffic to the target group instead of directly to individual instances.
- **Health Checks:** ELB regularly checks the health of targets using a path like /health. If a target becomes unhealthy, ELB stops routing traffic to it until it recovers.
- **Cross-Zone Load Balancing:** Distributes traffic evenly across healthy targets in all enabled Availability Zones, helping prevent overload in a single zone.

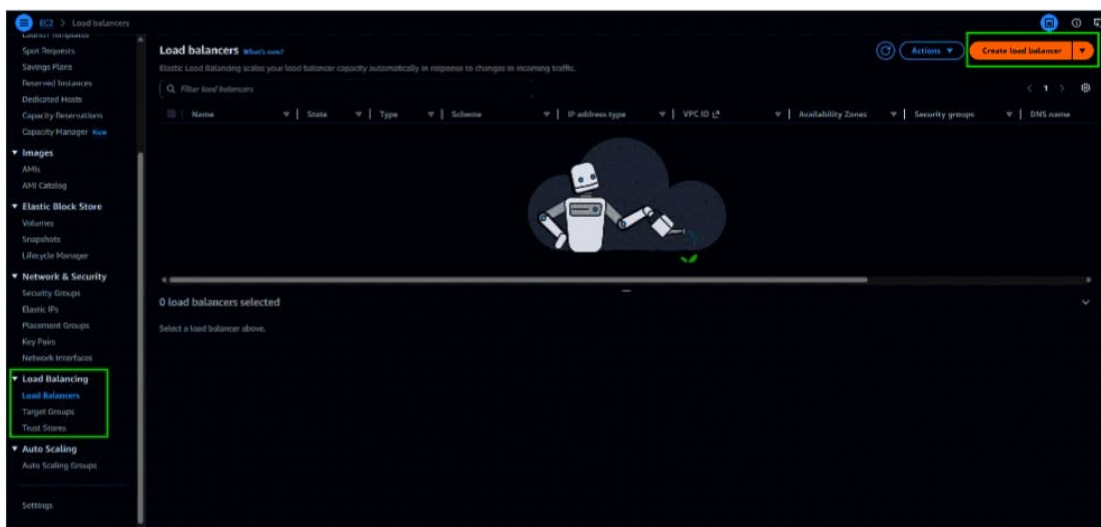


Steps to configure an Application Load Balancer in AWS

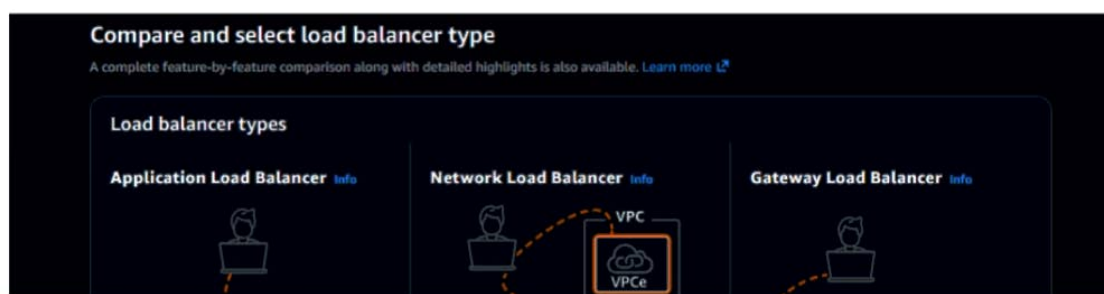
Step 1: Launch the two instances on the AWS management console named Instance A and Instance B. Go to services and select the load balancer. To create AWS free tier account refer to [Amazon Web Services \(AWS\) – Free Tier Account Set up](#).

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4	Elastic IP	IPv6 IPs
InstanceA	i-0e8b79e8280a5c8d1	Running	t3.micro	Initializing	ap-south-1a	ec2-45-1-84-174.ap-south-1a.amazonaws.com	65.1.84.174		
InstanceB	i-05278f10b593ba64a	Running	t3.micro	Initializing	ap-south-1a	ec2-52-46-200-110.ap-south-1a.amazonaws.com	52.66.200.110		

Step 2: Scroll down on the left side panel, go to Load Balancers, and click on Create Load Balancer.



Step 3: Select Application Load Balancer and click on Create.



Step 3: Select Application Load Balancer and click on Create.

Compare and select load balancer type

A complete feature-by-feature comparison along with detailed highlights is also available. [Learn more](#)

Load balancer types

Application Load Balancer [Info](#)

Choose an Application Load Balancer when you need a flexible feature set for your applications with HTTP and HTTPS traffic. Operating at the request level, Application Load Balancers provide advanced routing and visibility features targeted at application architectures, including microservices and containers.

[Create](#)

Network Load Balancer [Info](#)

Choose a Network Load Balancer when you need ultra-high performance, TLS offloading at scale, centralized certificate deployment, support for UDP, and static IP addresses for your applications. Operating at the connection level, Network Load Balancers are capable of handling millions of requests per second securely while maintaining ultra-low latencies.

[Create](#)

Gateway Load Balancer [Info](#)

Choose a Gateway Load Balancer when you need to deploy and manage a fleet of third-party virtual appliances that support GENEVE. These appliances enable you to improve security, compliance, and policy controls.

[Create](#)

▶ **Classic Load Balancer - previous generation**

[Close](#)

Step 4: Here you are required to configure the load balancer. Write the name of the load balancer. Choose the scheme as internet facing.

Basic configuration

Load balancer name
Name must be unique within your AWS account and can't be changed after the load balancer is created.

my-loadbalancer

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme [Info](#)
Scheme can't be changed after the load balancer is created.

☒ **Internet-facing**

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ **Internal**

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type [Info](#)
Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

☒ **IPv4**

Includes only IPv4 addresses.

☐ **Dualstack**

Includes IPv4 and IPv6 addresses.

☐ **Dualstack without public IPv4**

Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with Internet-facing load balancers only.

Step 5: Add at least 2 availability zones. Select us-east-1a and us-east-1b

Network mapping

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC [Info](#)
The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-06324974bb13f3388 (default) [Create VPC](#)

IP pools [Info](#)
You can optionally choose to configure an IPAM pool as the preferred source for your load balancer IP addresses. Create or view Pools in the [Amazon VPC IP Address Manager console](#).

☐ **Use IPAM pool for public IPv4 addresses**

The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets [Info](#)
Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ **ap-south-1a (aps1-a-1)**

Subnets

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL, inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.

subnet-0a8e5d0a61fcbad4
172.31.12.0/20
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (/)

☒ **ap-south-1b (aps1-a-3)**

Subnets

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL, inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.

subnet-0155c269fcc37e55
172.31.0.0/20
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (/)

☐ **ap-south-1c (aps1-a-2)**

Step 6: Select the default security group.

Security groups

A security group is a set of firewall rules that control the traffic to your load balancer. Listed security groups are in the same VPC as you selected above. Selected security groups must include at least 1 inbound rule and 1 outbound rule to allow traffic to your load balancer.

Security groups

Select up to 5 security groups

default (sg-0bde4b54dec2b140d) [Create security group](#)

Inbound rule (1) Outbound rule (1)

Step 7: In the Listener and Routing section, click on Create New Target Group.

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Listeners and routing [info](#)

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its **registered** targets.

Listener HTTP80 Remove

Protocol: HTTP Port: 80 1-65535

Default action [info](#)

The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action

☒ Forward to target groups ☐ Redirect to URL ☐ Return fixed response

Forward to target group [info](#)

Choose a target group and specify routing weight. [Create target group](#)

Target group: Weight: 1 0-999 Percent: 100%

+ Add target group

You can add up to 4 more target groups.

Target group stickiness [info](#)

Enables the load balancer to bind a user's session to a specific target group. To use stickiness the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.

☐ Turn on target group stickiness

Listener tags - optional

Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

[Add listener tag](#)

You can add up to 50 more tags.

Step 8: Choose the Target Group Name, select the Protocol, and enter the Port Number.

Step 1 Create target group

Step 2 - recommended Register targets

Step 3 Review and create

Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Settings - immutable

Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.*

Target type

☒ **Instances**
Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.
Suitable for: ALB, NLB, GWLB

☐ **IP addresses**
Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.
Suitable for: ALB, NLB, GWLB

☐ **Lambda function**
Supports load balancing to a single Lambda function. ALB required as traffic source.
Suitable for: ALB

☐ **Application Load Balancer**
Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.
Suitable for: NLB

Target group name

Name must be unique per Region per AWS account.
my-target-group

Accepts alphanumeric characters and hyphens. 1-32 total characters. Count: 15/32

Protocol

Protocol for communication between the load balancer and targets.
HTTP

Port

Port number where targets receive traffic. Can be overridden for individual targets during registration.
80 1-65535

IP address type

Only targets with the indicated IP address type can be registered to this target group.

☒ **IPv4**
Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

☐ **IPv6**
Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

Step 9: Choose instance A and instance B and Click on Next: Review.

Register targets - recommended

This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (2/2)

Instance ID	Name	State	Security groups	Zone	Private IPv4 address	Subnet ID	Launch
i-05279f10c593a64a	InstanceB	Running	default	ap-south-1a	172.31.39.226	subnet-0a4e5d0a1fcbad4	May 25
i-0e0b79e0280f05c0d	InstanceA	Running	default	ap-south-1a	172.31.44.214	subnet-0a4e5d0a1fcbad4	May 25

2 selected

Ports for the selected instances

Ports for routing traffic to the selected instances.
80 1-65535 (separate multiple ports with commas)
[Include as pending below](#)

Review targets

Targets (0)

[Show only pending](#)

Instance ID	Name	Port	State	Security groups	Zone	Private IPv4 address	Subnet ID	Launch time
No instances added yet								
Specify instances above, or leave the group empty if you prefer to add targets later.								

0 pending

[Cancel](#) [Previous](#) [Next](#)

Step 10: Review all the configurations and click on Create Target Group. After that, return to the Load Balancer Creation page and select my-target-group in the routing section.

Step 1 Create target group

Step 2 - recommended Register targets

Step 3 Review and create

Review and create

Review your target group configuration before creating.

Step 1: Target group details [Edit](#)

Target group details

Name: my-target-group	Target type: Instance	Protocol : Port: HTTP: 80	Protocol version: HTTP1
VPC: vpc-06324974bb13f3388	IP address type: IPv4		

Health check details

Health check protocol: HTTP	Health check path: /	Health check port: traffic-port	Interval: 30 seconds
Timeout: 5 seconds	Healthy threshold: 5	Unhealthy threshold: 2	Success codes: 200

Step 10: Review all the configurations and click on Target Group. After that, return to the Load Balancer Creation page and select my-target-group in the routing section.

Step 11: Scroll down and click on Create Load Balancer.

Step 12: Congratulations!! You have successfully created a load balancer.

Name	Status	Type	Scheme	IP address type	VPC ID	Availability Zones	Security groups	DNS name	ARN
my-loadbalancer	Active	application	Internet-facing	IPv4	vpc-06324974db13f3388	2 Availability Zones	sg-0b4e6c4d8c20314	my-loadbalancer-13303396...	ARN

Step 13: This highlighted part is the DNS name which when copied in the URL will host the application and will distribute the incoming traffic efficiently between the two instances.

Step 14: This is the listener port 80 which listens to all the incoming requests

Protocol/Port	Default action	Rules	ARN	Security policy	Default SSL/TLS certificate	mTLS	Trust store	Trust store assoc.
HTTP/80	Forward to target group: my-target-group (1 (HTTP))	1 rule	ARN	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable

Step 15: This is the target group that we have created

Name	ARN	Port	Protocol	Target type	Load balancer	VPC ID
my-target-group	arn:aws:elasticloadbalancing:ap-south-1:1330339685:loadbalancer/targetgroup/my-target-group	80	HTTP	Instance	my-loadbalancer	vpc-06324974db13f3388

Step 16: Now we need to delete the Load balancer. Go to Actions -> Click on Delete.

Step 17: Also don't forget to terminate the instances.

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4	ELB
Instance1	i-0a0b0b0b0b0b0b0b0	Running	t3.micro	OK	ap-south-1a	ec2-40-1-84-174.ap-south-1	52.88.200.110	ELB
Instance2	i-06279919c305a044a	Running	t3.micro	OK	ap-south-1a	ec2-52-46-200-110.ap-south-1	52.88.200.110	ELB